

Module E: a recombination-matched haplodiploid neutral null for ancestry sorting in replicate *Formica* hybrid populations

Abstract

Modules A–C established that parental ancestry sorts in a directional, partly predictable way across 20 replicate *F. polyctena* $\times$ *F. aquilonia* hybrid populations, but those results are descriptive: without a neutral null they cannot distinguish selection from drift. Module E builds that null. Founders are seeded from *real phased parental haplotypes* rather than independent allele-frequency draws, so within-species linkage disequilibrium is preserved; recombination follows the empirical genetic map; the model is haplodiploid; and all selection is removed, leaving drift and recombination as the only forces. The key design element is an *internal control*: markers are stratified by DiagnosticIndex (DI), so low-DI markers—which cannot be driven by ancestry sorting—calibrate the demography within the same runs that test the high-DI markers. Observed among-population differentiation rises steeply with DI ($F_{ST} = 0.051 \rightarrow 0.315$) while the neutral null is essentially flat ($0.191 \rightarrow 0.156$), and the fraction of sorted loci exceeds the null by ~ 1.5 -fold at low DI rising to >100 -fold at high DI. Because drift is DI-blind by construction, a DI-scaled excess cannot be produced by demographic misspecification. The null does not yet reproduce the neutral background itself (it over-differentiates $3\text{--}4\times$ at $\text{DI} \leq -90$), indicating a simulated effective size that is too small; recalibrating on these neutral markers—rather than on the ancestry-informative markers, which are the test—lowers simulated differentiation in every DI bin and thus *widens* the high-DI excess, so the reported magnitudes are conservative.

1 Rationale

Near-fixation of a parental allele in a hybrid population is expected under a Dobzhansky–Muller incompatibility or consistent directional selection, but also under plain drift in small, young populations. The two are separated by *direction*: selection predicts the same parent fixing repeatedly across replicates, drift predicts random directions. Quantifying “more parallel than drift” requires a null that reproduces the actual demography, recombination landscape and sampling design of the data. That is what Module E provides.

2 Methods

2.1 Conditional design

Rather than simulating the parental species’ history, we *condition* on it. Founders are drawn from real phased parental haplotypes, the mutation rate is set to zero, and only the post-contact hybrid phase is simulated. This inherits true allele frequencies, true between-species divergence and—critically—true within-species haplotype structure, none of which independent binomial draws at empirical frequencies would reproduce.

2.2 Founder haplotypes

Fifteen diploid females per species were imputed and phased *within species* with Beagle 5.4, informed by the empirical cM map (phasing across the species boundary would manufacture artificial intermediate haplotypes). Two *polyctena* individuals with $>40\%$ missing data were excluded, giving a founder pool of 30 *aquilonia* haplotypes and 13 *polyctena* diploid females over

1,114,251 markers. Phasing preserved the pool faithfully: correlation between raw and phased allele frequencies $r = 0.999$, expected heterozygosity 0.216/0.203 against raw 0.216/0.206, and mean between-species $|\Delta p|$ of 0.304 against 0.303.

2.3 Simulation model

SLiM 5, non-Wright–Fisher, haplodiploid with arrhenotoky: haploid males from unfertilised eggs, diploid females, spermathecal sperm storage, multiple mating following Pamilo, and an age-structured life table. The genome is the 26 assembled chromosomes (chromosome 23 excluded), each with its empirical recombination map. Maps are in cM/Mb and must be converted with `scale=10-8` to per-base rates. Mutation rate is zero, so the simulation redistributes existing real variation rather than generating new variation. All selection (QTN, DMI, climate) is removed.

Founders are read from one combined per-chromosome VCF containing 30 haploid *aquilonia* samples and 13 diploid *polycytena* samples. A single read call per chromosome is required: reading the two species separately creates two distinct mutations at each position instead of one shared biallelic marker. Each parameter set is run as 20 independent demes—each an independent founder draw and independent drift history—sampled every 20 generations to generation 160.

2.4 Marker sets and DI stratification

Two marker sets were used. The first is the ancestry-diagnostic set, $DI > -25$ (49,506 markers). The second, used for the results reported here, takes the locked Module A gate (pooled parental $MAF \geq 0.15$; 650,950 markers, DI from -168 to -4) and draws **4,000 markers from each of 10 fixed DI bins** (40,000 total). Equal allocation per bin is deliberate: uniform thinning is swamped by the mid-DI bulk and starves both tails.

This stratification supplies the internal control. Low-DI markers carry little ancestry information, so ancestry sorting cannot drive their statistics; they calibrate the demography *in the same runs* that test the high-DI markers. The test is then the *shape* of the discrepancy across DI: a gap that rises with DI implicates ancestry-specific processes, whereas a flat gap implicates demographic misspecification.

2.5 Requirements for a valid comparison

Three points proved essential and are worth recording.

1. **Sample-size matching.** Composite r^2 carries a finite-sample bias of order $1/2n$, and `sort_class` likewise fires more readily in small samples (a 3-diploid sample can near-fix trivially; 50 diploids essentially cannot). Simulated demes are therefore subsampled to the empirical per-population sizes (3–20 diploids). Without this the null appears artificially unsorted and the LD comparison is biased toward younger ages.
2. **Symmetric treatment.** Any filter applied to the data must be applied to the null. “Sorted” is by definition high among-population differentiation, so removing sorted loci from the observed data alone mechanically lowers its F_{ST} and moves it toward the null irrespective of cause.
3. **Shared orientation.** Both datasets are polarised with the same parental reference (the phased founder pool) and the locked Module A conventions ($MAF \geq 0.15$, `fix_th` = 0.15, `sort_th` = 0.5, `null_prob` = 0.5).

3 Controls excluding neutral alternatives

Before interpreting any excess, several neutral explanations were quantified and excluded.

Founder heterogeneity. If replicate populations were founded from different local source populations, among-population differentiation would be inflated neutrally. Splitting the parental pool into k balanced sub-pools and founding each deme from one of them gives an at-founding among-deme F_{ST} of only 0.0095–0.024 ($k = 2$ –4). The absolute ceiling—each deme founded from a single *aquilonia* haplotype and a single *polycтена* female, the most extreme founding available—is 0.090. The mechanism is general: at ancestry-informative markers both species are near-fixed for opposite alleles, so any *aquilonia* sub-pool resembles “aquilonia” and any *polycтена* sub-pool “polycтена”; a 50/50 admixture washes out within-species structure regardless of how divergent the sources are.

Colony (nest) structure. The empirical samples are workers from few colonies (164 workers from 39 nests; median 2 nests per population; 9 of 20 populations are a single colony), and nestmates could in principle be siblings, which would depress π and inflate F_{ST} . Measured directly, the excess within-nest pairwise genotype correlation is only 0.047 (0.565 within nests versus 0.517 between nests in the same population): nestmates are barely more related than non-nestmates, as expected for polygynous supercolonial *Formica*. The effect is real but minor.

Among-population ancestry variance. Populations differ in mean ancestry (SD 0.065, range 0.40–0.66), which contributes $\text{Var}(a)/\bar{p}(1 - \bar{p}) \approx 0.017$ to F_{ST} at diagnostic markers—about 11% of the observed high-DI gap.

Individual outliers. Sielva is a near- F_1 population (observed heterozygosity 0.813 against 0.499 expected from its ancestry; within-population ancestry SD 0.001), and is genuinely young. Removing it changes π from 0.299 to 0.293 and F_{ST} from 0.313 to 0.320; across all 20 leave-one-out analyses the largest change in F_{ST} is 0.009. No single population drives the result.

4 Results

4.1 The neutral null is anchored on markers that cannot sort

Low-DI markers ($\text{DI} \leq -60$) were used to fit the demography by LD decay, selecting $K = 1500$ at generation 160 (Fig. 2). This anchor must be computed *within* populations and averaged; pooling individuals across differentiated populations manufactures LD and destroys the comparison.

4.2 Differentiation scales with DI in the data but not under drift

Table 1 and Fig. 1 give the central result. Observed F_{ST} rises from 0.051 at the neutral background to 0.315 at the most ancestry-informative markers, while the neutral null is nearly flat (0.191 $\rightarrow$ 0.156). The fraction of sorted loci shows the same pattern more sharply, from a ~ 1.5 -fold excess at low DI to >100 -fold at high DI.

4.3 Why the shape matters

Drift does not distinguish ancestry-informative from uninformative markers: a purely demographic error—wrong effective size, wrong age, wrong founding size—shifts all DI bins together. The simulated profile confirms this, remaining flat across the full DI range. A discrepancy that *scales* with ancestry-informativeness therefore cannot be produced by misspecifying the demography, and the comparison is internally controlled because both datasets use the same markers, the same MAF gate and the same frequency spectrum.

Table 1: Observed and simulated statistics by DI bin at the anchor-matched neutral cell ($K = 1500$, generation 160). “sorted” is the fraction of loci classified as directionally sorted.

DI bin	F_{ST} obs	F_{ST} null	π obs	π null	sorted obs	sorted null
≤ -90	0.051	0.191	0.448	0.381	0.026	0.021
$(-90, -75]$	0.257	0.185	0.308	0.344	0.184	0.103
$(-75, -60]$	0.303	0.183	0.260	0.314	0.350	0.233
$(-60, -50]$	0.307	0.179	0.251	0.313	0.398	0.267
$(-50, -40]$	0.309	0.172	0.253	0.328	0.378	0.183
$(-40, -30]$	0.305	0.167	0.277	0.364	0.260	0.072
$(-30, -25]$	0.312	0.161	0.293	0.388	0.175	0.030
$(-25, -20]$	0.310	0.160	0.297	0.398	0.150	0.015
$(-20, -15]$	0.314	0.161	0.302	0.405	0.130	0.001
> -15	0.315	0.156	0.304	0.415	0.116	0.000

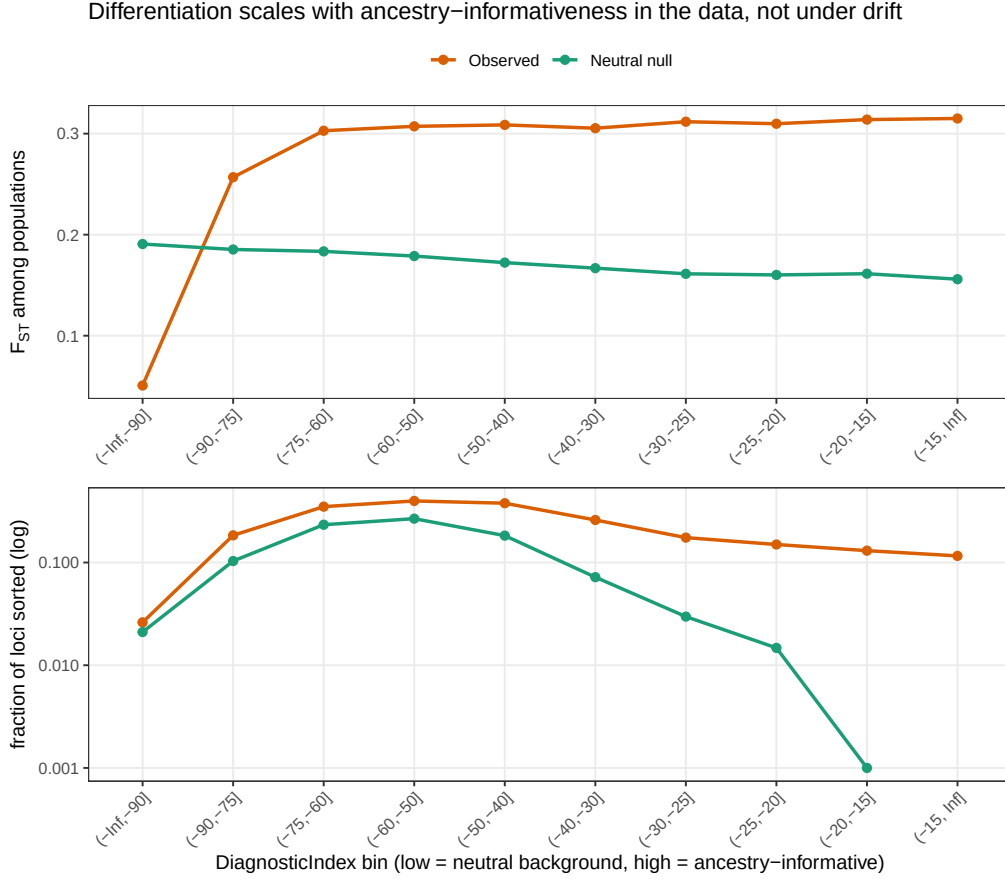


Figure 1: Among-population differentiation (top) and the fraction of sorted loci (bottom, log scale) against DiagnosticIndex bin, for the observed data and the neutral null. Neutral drift is DI-blind and produces an essentially flat profile; the data concentrate both differentiation and sorting in ancestry-informative markers.

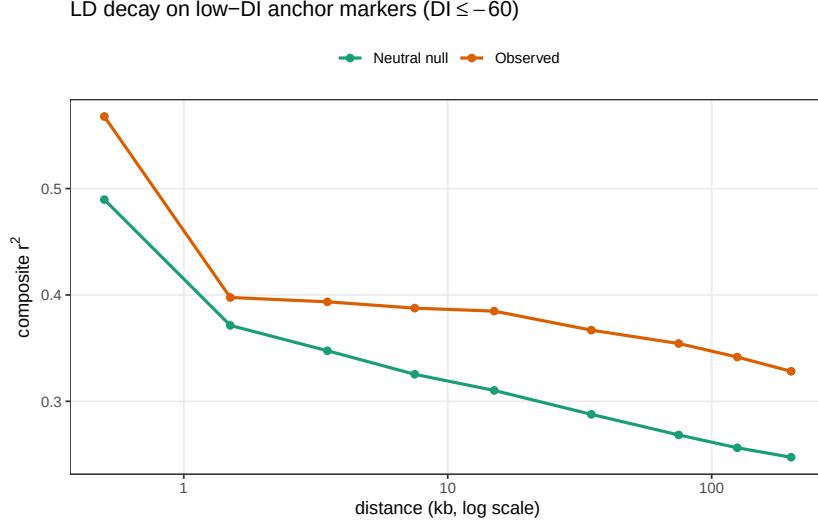


Figure 2: LD decay on the low-DI anchor markers ($DI \leq -60$), computed within populations and averaged, for the observed data and the anchor-matched neutral cell.

5 Limitations and the planned calibration

The null does not yet reproduce the neutral background. At $DI \leq -90$ the observed F_{ST} is 0.051 while the null gives 0.14–0.19 (depending on K): the simulation over-differentiates by $3\text{--}4\times$ on markers where sorting cannot be the cause, and simultaneously undershoots neutral π (~ 0.40 against 0.448). Both signal the same thing—the *simulated effective size is too small*: isolated demes drift apart more, and lose more within-population diversity, than the real populations do at neutral loci. Consequently the *magnitude* of the excess is provisional, though its *direction* is not: reducing simulated drift lowers F_{ST} in every DI bin and, because the null is flat across DI, *widens* the high-DI gap, making the reported values conservative.

Planned analysis. The calibration must be performed *on the low-DI markers*—the internal control that cannot sort—rather than on the ancestry-informative markers (which are the test) or on LD decay (which, being generated partly by drift, favours the opposite direction and disagrees with π/F_{ST}). Two demographic parameters are swept over the full DI-stratified marker set:

1. **carrying capacity** K , extended *upward* beyond the values run so far (the neutral background requires less drift than even $K = 6250$ provides, so larger effective sizes are needed);
2. **number of founders** per deme, which sets both within-population diversity and among-deme founding variance.

The cell that reproduces neutral F_{ST} (≈ 0.051) and π (≈ 0.448) then defines the calibrated null, and the high-DI excess is read off there. Because the null’s F_{ST} is flat across DI, matching the neutral background at ~ 0.05 lowers the simulated high-DI F_{ST} to the same level, so the observed high-DI value (0.315) stands against a much lower neutral expectation than in Table 1; the dose-response is expected to strengthen, not weaken, under this recalibration. Founding is expected to be constrained to be *large*: fewer founders lower π and raise F_{ST} , moving the neutral background away from the observed values, which itself excludes a tight-bottleneck neutral explanation on the markers where the test is trustworthy.

Until the background is reproduced, the honest statement is that **differentiation and sorting scale with ancestry-informativeness in a way neutral drift does not produce**, with the magnitude of that excess to be fixed once the null reproduces the markers that cannot sort.